

SAQIB ALI

Electrical Engineering Student · Motor Drives, Control Systems & Satellite Ground Stations

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EDUCATION

B.Tech, Electrical Engineering

Zakir Husain College of Engineering & Technology, AMU

Sep 2023 – Jun 2027

Aligarh, Uttar Pradesh

EXPERIENCE

Research Intern

Indian Institute of Technology Ropar

Jun 2026 – Jul 2026

Rupnagar, Punjab

- Modelled a five-phase IPMSM in MATLAB/Simulink and designed dual-plane field-oriented control with MTPA and voltage-limited flux weakening.
- Derived a loss-aware MTPA strategy using Lagrange multipliers with a Newton–Raphson solver; implemented direct torque control and five-phase SVPWM.
- Generated ePWM signals on a TI C2000 DSP using Code Composer Studio and validated the gate driver circuit on hardware.

Lead Engineer

Orbitalink Private Limited

Aug 2025 – Present

Aligarh, Uttar Pradesh

- Lead end-to-end development of a two-axis satellite antenna tracker, from the Fusion 360 chassis and RF front end to motion firmware.
- Designed a 435 MHz cross-Yagi antenna in MMANA-GAL, tuning the feedpoint to 36.4Ω with a two-stage phasing harness for circular polarisation.
- Wrote firmware driving NEMA23 hybrid servo motors via HSD57 drivers with trapezoidal motion profiles, EEPROM-stored parameters and a serial command protocol; porting it to STM32F411 as modular code.
- Manage field-unit installation and cabling, and debug hardware–software integration issues.

Electrical Design Intern

Orbitalink Private Limited, with AMU Innovation Foundation

2025

Aligarh, Uttar Pradesh

- Designed an adaptive PID controller to stabilise and precisely point a high-inertia antenna tracking mount.

Ground Station Team Member

SS AMU SAT, University Satellite Programme

Sep 2024 – Present

Aligarh, Uttar Pradesh

- Built the tracker chassis CAD model and core two-axis PID positioning logic; support satellite tracking, equipment calibration and signal reception.

PROJECTS

Image Downlink over Software-Defined Radio | CM4, Python, GNU Radio

- Wavelet-based image compression on a Raspberry Pi Compute Module 4, transmitted over radio with a GNU Radio flowgraph.

DC-DC Boost Converter Modelling | Simulink, Power Electronics

- Simulated steady-state and transient response across duty cycles and load conditions.

Weather Dashboard | Node.js, Axios, OpenWeatherMap API

- Responsive web app with live worldwide weather data, search and API error handling; deployed on Render.

TECHNICAL SKILLS

Control & Power: FOC, MTPA, flux weakening, DTC, SVPWM, adaptive PID, system modelling, DC-DC converters

Embedded: C/C++, TI C2000 (CCS), STM32 (CubeIDE), Arduino, I2C, sensor interfacing, stepper/servo drives

RF & Design: Yagi antenna design, MMANA-GAL/NEC, GNU Radio, Fusion 360, finite element analysis

Software: MATLAB/Simulink, Python (NumPy, Pandas, scikit-learn), JavaScript, Node.js, Git

CERTIFICATIONS

Electrical Engineering Job Simulation, GE Aerospace | MATLAB Simulation and Control, IEEE SB AMU (2025)
Advanced Aircraft Design Workshop, Bharat Space Education Research Centre (2024)